

Topic: Nested Queries (Subqueries)

Q. What are Nested Queries (Subqueries) in SQL? Differentiate between Single-Row, Multiple-Row, and Correlated subqueries with examples.

📌 Definition to Remember

A **Nested Query** (or **Subquery**) is a complete SQL query placed **inside another SQL query** (the outer query). The inner query executes first and provides data to the outer query. Subqueries can be placed in `WHERE`, `HAVING`, or `FROM` clauses.

1. Types of Subqueries

A. Single-Row Subquery

- Returns exactly **one row and one column** to the outer query.
- Uses single-value comparison operators: `=`, `>`, `<`, `≥`, `≤`, `≠`.

```
-- Find employees who earn more than the average salary
SELECT Name, Salary FROM Employee
WHERE Salary > (SELECT AVG(Salary) FROM Employee);
```

The inner query returns one value (e.g., 45000). The outer query uses it.

B. Multiple-Row Subquery

- Returns **more than one row** to the outer query.
- Must use multi-row operators: `IN`, `ANY`, `ALL` (not `=`).

```
-- Find employees in departments located in 'New York'
SELECT Name FROM Employee
WHERE Dept_ID IN (SELECT Dept_ID FROM Department WHERE City = 'New York');
```

C. Correlated Subquery

- The inner query **references a column from the outer query** — they are linked.
- The inner query is **re-executed for every single row** of the outer query.
- More powerful but **computationally expensive**.

```
-- Find employees earning above their OWN department's average salary
SELECT e1.Name, e1.Salary, e1.Dept_ID
FROM Employee e1
WHERE e1.Salary > (
    SELECT AVG(Salary)
    FROM Employee e2
    WHERE e2.Dept_ID = e1.Dept_ID -- Correlation: ties inner to outer
);
```

2. Comparison of Subquery Types

Feature	Single-Row	Multiple-Row	Correlated
Rows Returned	Exactly 1	More than 1	Varies per outer row
Operators	= , > , <	IN , ANY , ALL	= , > , <
Execution	Once	Once	Once per outer row
Performance	Fast	Moderate	Slow (re-executes per row)

★ Must-Write Points (for 10 marks)

1. A Subquery/Nested Query is a SELECT query placed inside another SQL query.
2. The **inner query executes first** and provides data to the outer query.
3. Single-Row Subquery: returns one value; uses operators like = , > , < .
4. Multiple-Row Subquery: returns multiple rows; must use IN , ANY , or ALL .
5. Correlated Subquery: inner query references outer query — re-runs for every outer row.
6. Correlated subqueries are more powerful but slower (expensive execution).
7. Subqueries can be placed in WHERE , HAVING , or FROM clauses.

⚡ Quick Recall

Subquery (inner executes first) → Single-Row (=,>,< operators) → Multi-Row (IN/ANY/ALL) → Correlated (references outer, re-runs per row, expensive)